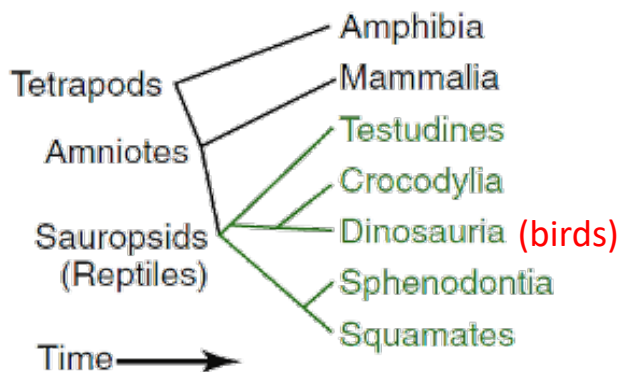


Clinical Anatomy and Physiology of Reptiles

Hugues Beaufre, DVM, PhD, DACZM, DABVP (Avian), DECZM (Avian)

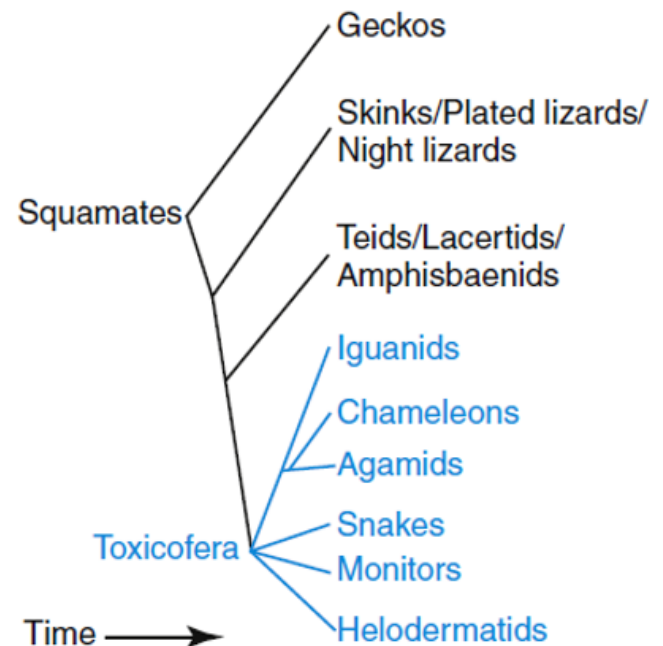
INTRODUCTION

The Class Reptilia is very diverse and include more than **10,000 species**. This class is **not a monophyletic class** (so it is paraphyletic), but, when including birds (clade of Sauropsida), it is monophyletic. For this reason, it is not uncommon in the scientific literature to refer to reptiles as **non-avian reptiles**. While birds look very different from reptiles and their captive management and medicine are different, **chelonians and crocodilians are phylogenetically closer to birds than they are to other reptiles**. As such, many anatomical and physiological features are similar between birds and reptiles, especially with crocodilians and somewhat in chelonians.



Reptiles comprise 4 orders. The Order **Rhynchocephalia** (also known as Sphenodontia) is only composed of a single species, the tuatara (*Sphenodon punctatus*), a lizard-like New Zealand reptile (you will not see it in captivity). The Order **Testudines** (also known as chelonians) includes about 300 species. This order is classically divided into two suborders, the **pleurodira** (side-necked turtles, all are freshwater turtles, they cannot retract their neck inside their shell) and the **cryptodira** (classic turtle and tortoise, can retract their head inside their shell). The cryptodira includes familiar

families such as the sea turtles (belonging to 2 families), fresh water turtles (most are in the **Emyridae** family, but other families as well), and the tortoises (all are in the **Testudinidae** family). While in the (American) English language the term tortoise refers to terrestrial chelonians, the term turtle to aquatic chelonians, and the term terrapin to semi-aquatic chelonians, they do not actually have much taxonomic or biological relevance (in lots of other languages for instance, there is a single word for all, like in French or Spanish). **The Order Squamata is the most diverse of all reptiles and include most of the species with more than 9,000 species.**



It is composed of all the lizards and snakes. While snakes look very different from lizards, they are actually one group of advanced lizards (suborder Serpentes) that are phylogenetically closer to some lizards than anticipated. For instance, a snake is phylogenetically closer to a bearded dragon than a bearded dragon is close to a gecko. Interestingly, snakes evolved from fossorial lizards that lost their limbs and completely lost their eyes in the process. When some species adapt-

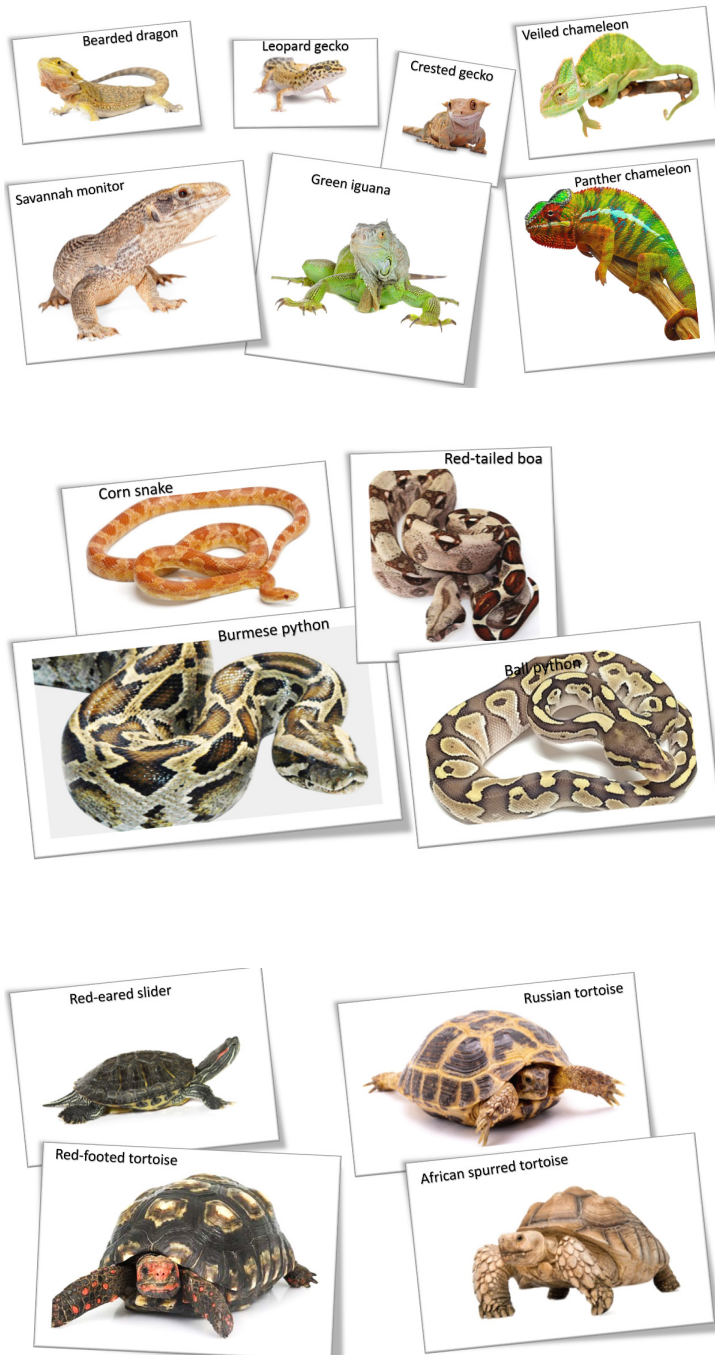
ed to surface life again, they re-evolved an eye, which explains why the snake eye is embryologically different from other reptiles. **Most snake species belong to the family Colubridae. Pythons (family Pythonidae) and boas (family Boidae) are also commonly seen as pets.** Venomous snakes include members of Viperidae (vipers, rattlesnakes), Elapidae (mambas, cobras, sea snakes), Colubridae (only a few species), and Atractaspidae (mole vipers).

The last order is the **Crocodylians**, which include gharials, crocodiles, and alligators. They are very close to

birds (crocodilians + birds = archosaurs). Their anatomy and physiology is very different from other reptiles. We will not cover this order in this lecture as they are uncommonly kept as pets, but will only highlight some interesting facts.

Please see below some reptile species of veterinary importance (table below only shows selected taxonomic units). The most common species that we see, by far, is the bearded dragon.

Order	Descriptive groups	Family	Species (examples)	
Testudines	Freshwater turtles	Emydidae	Western pond turtle Red-eared slider Box turtle	
		Chelydridae	Common snapping turtle Alligator snapping turtle	
		6 other families	Soft-shell turtles Sideneck turtles	
	Sea turtles	Cheloniidae	Green sea turtle	
		Dermochelyidae	Leatherback sea tuttle	
	Tortoises	Testudinidae	Sulcata tortoise Greek tortoise	
	Squamata (suborder: Lacertilia)	Old-World lizards	Agamidae	Bearded dragons Water dragons
			Chameleonidae	Veiled chameleon Panther chameleon
			Lacertidae	Typical wall/rock lizards
Scincidae			Skinks	
		Varanidae	Monitor lizards	
		New-World lizards	Iguanidae	Green iguana
			Teiidae	Argentinian tegu
Geckos		Gekkonidae	Leopard gecko Crested gecko	
		Many others	About 20 other families	
Squamata (suborder: Serpentes)		Colubrid	Colubridae	Cornsnake, kingsnake Garter snake Hognose snake
	Large constricting snakes	Pythonidae	Ball python Burmese python	
		Boidae	Boa constrictor (e.g. red-tailed boa)	
	Venomous snakes	Viperidae	Adders, vipers Rattlesnakes	
			Elapidae	Mambas, cobras Sea snakes
	Many others	About 20 other families		



1/7th of mammals. This is important to realize, as this is very clinically relevant for things like pharmacology, response to treatment, wound and bone healing, and anesthesia. **The amount of food necessary to sustain a mammal for a day may be enough for a week in reptiles.** The metabolism obviously varies per species with some slower (e.g. tuatara) and some faster (e.g. monitor lizards) than others.

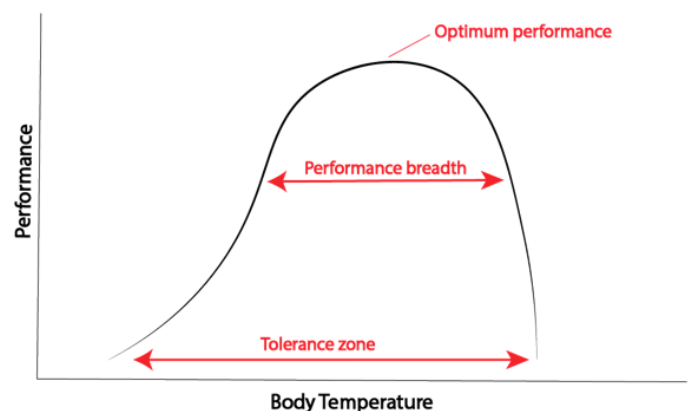
Another important feature of reptilian metabolism is that there is a limited capacity for aerobic metabolism. Indeed, reptiles switch to **anaerobic metabolism** frequently and with increased efforts, resulting in the build up of lactates. Consequently, most reptiles cannot have a sustained activity for very long (>5 minutes) without resting periods. The blood pH of reptiles will be much more labile than homeotherms and acidemia is common with disease, exertion, and apnea. Another consequence of this is that reptiles with respiratory diseases will only show clinical signs when the disease is advanced.

Finally, reptiles are poikilothermic (large variations in body temperature) and ectothermic (rely on environmental temperature, do not have abilities for thermogenesis). The main advantage of this strategy is that reptiles do not waste energy in thermogenesis and can thus thrive with limited energy sources. However, the main disadvantage is that all their activities are limited by ambient temperature. This means that their range will be restricted to warm and temperate climates and some species have to brumate to survive during the winter. With that said, there are a few exceptions with some reptiles with limited thermogenesis (the most notable one being the leatherback sea turtle). The term cold-blooded is misleading and should not be used as reptiles thrive to maintain a relatively high body temperature by using a variety of techniques. Each reptile species has a range of temperature within which they function at its physiological best; it is called the **preferred optimum temperature zone**.

This section will only cover the basics of anatomy and physiology of reptiles as there is not enough time to cover this topic more. Remember that most of the reptile internal anatomy is kind of a simplified version of birds without the air sacs (with some exceptions).

METABOLISM

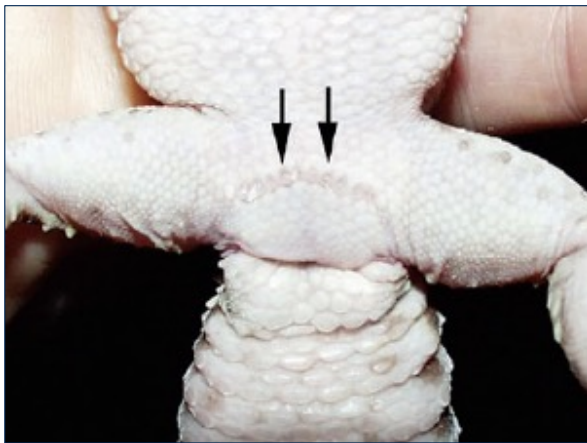
Reptiles have an overall **slower metabolism** than homeothermic vertebrates like birds and mammals. It is estimated that their metabolism is about 1/5th to



INTEGUMENT

The reptilian integument is obviously characterized by the presence of **scales**. While most species have well defined and conspicuous scales, some species do not have scales or have poorly developed scales (e.g. turtles, leopard gecko).

There are many different types of scales (non-overlapping, overlapping, with tactile sensory organ, with ridges, elongated). The scales of the geckos' toepad, have specialized surface made of millions of setae (hair-like structures) that help them [stick to walls](#). The reptile skin is composed of several layers (5 to be precise) with 2 different types of keratin: hard keratin or **β -keratin** (not found in mammals) and soft keratin or **α -keratin**. Scales are mainly made of β -keratin while the space in between, made of a thinner epidermis (the **hinges**), are mainly made of α -keratin. The scales may also have ornamental protrusions or create spines, horns, crest alone or in combination with osteoderms. In particular, **femoral pores or precloacal pores** in lizards can be used to sex them as they are more prominent in males.



Many reptiles have **osteoderms**, calcified dermal bones. They are well developed in crocodilians, in some lizards (e.g. *Uromastyx* spp.), and on the legs of some tortoises (e.g. African spurred tortoise).

Some lizard species have **chromatophores**, which

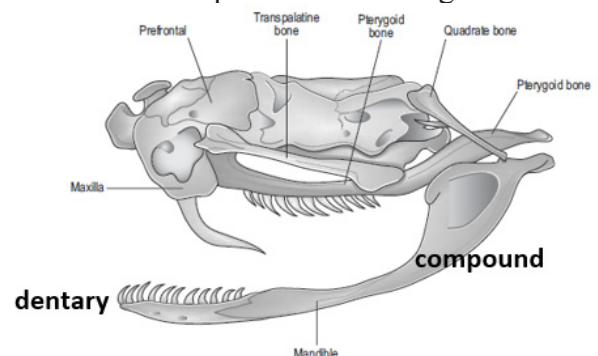
are cells in the dermis that contain pigments and allow some species to change colours (e.g. chameleons). (Note, these cells can tumorize and are thus called chromatophoromas or iridophoromas).

Reptiles shed their skin when growing and to replace worn skin. **Squamates shed their skin periodically either in one piece (snakes) or in several pieces (lizards)**. Chelonians and crocodilians shed their skin continuously or in tiny pieces. Most turtles shed their scutes unlike most tortoises. **The shedding process is also called ecdysis and shedding issues are called dysecdysis**. During the renewal phase of ecdysis, the germinal layer produces a new skin composed of the usual 5 layers and separates from the old skin by lymph.

SKELETAL SYSTEM

It is beyond the scope of this lecture to review the entire skeleton of reptiles, only specific features will be highlighted.

As in birds, **the reptilian skull is kinetic** with most species being able to raise the upper jaw. The skull is particularly kinetic in snakes with lots of ligamentous connections between bones. The upper jaws can move independently and literally walk around a prey when swallowing. The mandibles do not have a symphysis and can stretch laterally to widen the mouth opening and an additional joint is present in the middle of the mandible (the mandible is thus divided into the dentary bone rostrally and the compound bone caudally). The quadrate bone is the most moveable bone of the skull and links the lower to the upper jaw. Parts of the skull are not ossified in squamates including the braincase.

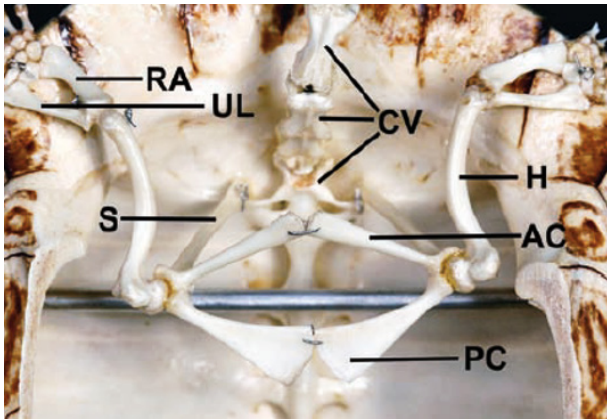


Jacobson

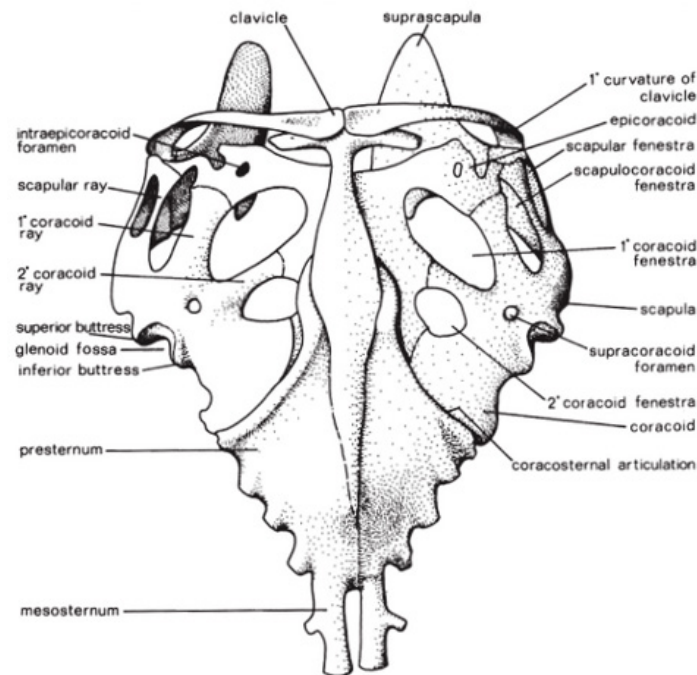
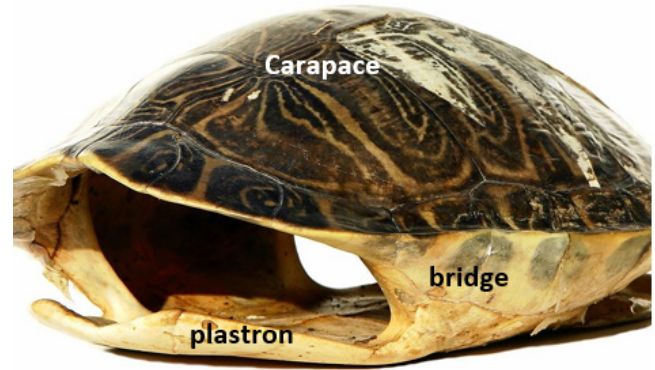
Within the skeleton, the pectoral girdle anatomy warrants a special mention as it has great significance when doing radiographs or CT-scans. In chelonians, parts of the bones have been incorporated into the plastron and the remaining bones form a tri-radiate structure connected to the glenoid articulation and formed by the

scapula, the acromion process of the scapula, and the procoracoid.

In lizards, we find a coracoid bone as well as an additional bone, the epicoracoid. The sternum is also more or less fenestrated depending on the species.



Jacobson

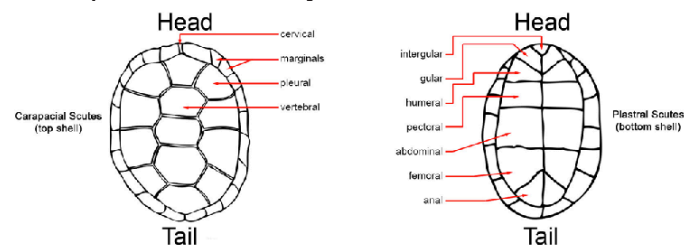


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The pectoral girdle is absent in snakes.

The shell is classically divided into the dorsal **carapace**, the ventral **plastron**, and the **bridges** linking the two laterally. The carapace originates from broadly expanded ribs and vertebrae fused with some dermal bones. The plastron has a different origin and comes from the fusion of dermal bones, including clavicles, interclavicle, and enlarged gastralia, replacing the typical sternum (chelonians do not have a sternum). The shell is covered by epidermal keratinized **scutes** that do not correspond to underlying dermal bones, and have a certain nomenclature used for topography.

Tail autotomy is the ability to self-amputate the tail along specific vertebral plans in many species of lizards as a **defense mechanism**. Lizards should not be restrained by their tail for this reason. The tail also tends to grow back (including with its spinal cord). Interestingly, agamidae (and therefore the bearded dragon) cannot perform autotomy.



COELOMIC CAVITIES

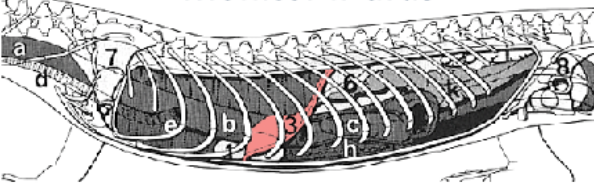
The partitions of the coelom are not homogeneous across reptiles. In chelonians, the **post-pulmonary septum** (this septum is split into two in birds by the air sacs and becomes the oblique and horizontal septa) is immediately ventral to the lungs and separate the coelom into the pleural cavities and the peritoneal cavity. In emydidae (freshwater turtles), the septum is incomplete caudally. In squamates, there are no separation in most species with a **single pleuroperitoneal cavity** (even the lungs are free floating in the coelom). Some exceptions exist notably in monitor lizards (varanidae) where the post-pulmonary septum divides the coelom

into the pleural and peritoneal cavity, much like in chelonians. In tegus (teiidae), a mid-body **post-hepatic septum** (also present in birds) partitions the coelom into the thoracic and abdominal cavities. Crocodilians have both the post-pulmonary and post-hepatic septum, much like in birds.

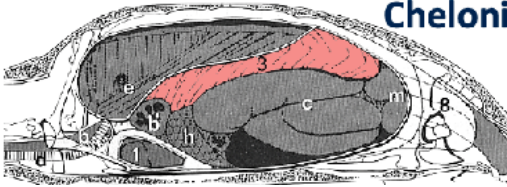
Tegus



Monitor lizards



Chelonians



GENERALITIES ON INTERNAL ANATOMY

Most reptiles seen in the clinics have most of their

internal organs in a single cavity, the pleuroperitoneal cavity (see exceptions above). Consequently, it is relatively easy to access most organs through coeliotomy incisions or coelioscopy.

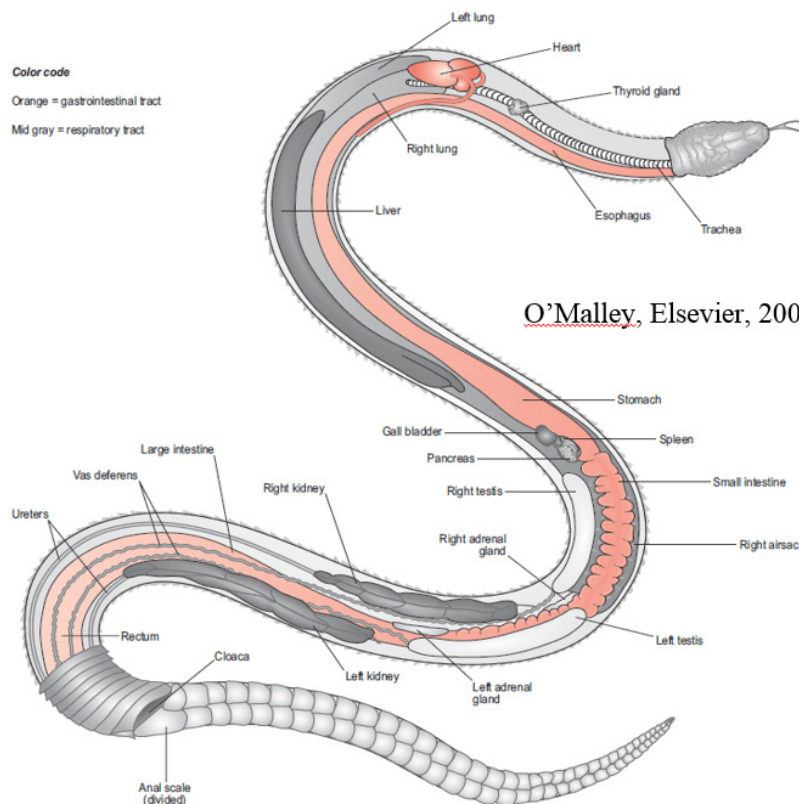
Snakes have an elongated anatomy and most of the organs have an elongated structure. In general, right organs are slightly more cranial than left organs and most species have a single lung (see later). The heart is very elongated and kind of twisted too.

Reptiles do not store fat in the skin and subcutaneous space. Most species have **2 fat bodies** within the peritoneal cavity that are palpable. Thus, obese reptiles will mainly have coelomic distention but their limbs and trunk would look relatively normal otherwise. Some species such as the leopard gecko store fat in the tail too.

DENTAL ANATOMY

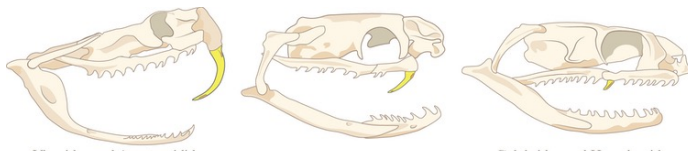
Dental anatomy varies across reptile families and deserves special mention. The **different types of dentitions** are outlined in the following table.

Snakes in particular have many teeth. Pythons and boas for instance have a total of 6 rows of teeth, 4 on the top and 2 on the bottom. The dentition of snakes as it pertains to venom teeth is also highly variable. When the venom teeth (fangs) are caudally located inside the mouth, the snake is said to be **opisthoglyphous** and it



Type	Description	Species	Picture
Acrodon	Teeth on top of the bone Not replaced in adulthood Acrodon species are more prone to periodontal disease (which is common)	Agamids (bearded dragons, uromastyx) Chameleons	
Pleurodon	Teeth on the inner margins of the bone Teeth replaced if lost during adulthood or periodically	Iguanids Snakes	
Thecodont	True dental alveoli	Crocodilians	

occurs mainly in venomous colubridae. When the fangs are rostral, it is called **proteroglyphous** and is found in most of the venomous snakes such as vipers, cobras, mambas, sea snakes and so on. Some species also have long fangs that can be folded against the roof of the mouth when the mouth is closed, these species are qualified as **solenoglyphous**, such as vipers.

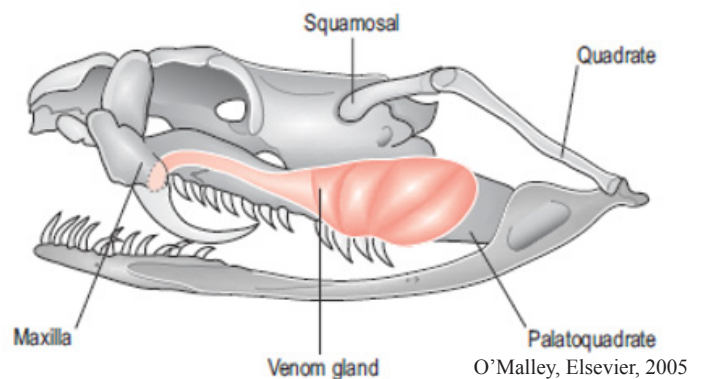


GASTROINTESTINAL SYSTEM

The anatomy of the GI depends obviously on the species and its natural diet. It tends to be short in carnivorous species such as crocodilians and snakes and long in more herbivorous species such as many lizards and tortoises. **Herbivorous reptiles tend to have a large colon where microbial fermentation occurs.** A cecum is present in some species, but most of the fermentation occurs in the colon. The stomach of snakes is very distensible and most snakes do not have a cecum (except in pythons).

Since we talked about the different types of fangs in snakes, maybe we can touch on their venom glands. Non-colubrid venomous snakes have a large venom

gland in the temporal area with a long duct that connects to the fang and discharge venom from a pressurized reservoir. In colubrids, the venom gland is at more or less the same location, but does not have a reservoir and does not discharge venom with pressure; it is called the **Duvernoy's gland**. As the venom duct is short, it is very difficult to remove it in these snakes whereas a venom surgery can be performed in the other species. In lizards, the helodermatid lizards are venomous (beaded lizard and gila monster) and their venom gland is along the jaw. (note: if you look at the phylogeny tree above, you will see that a whole clade of squamates is called the **toxicofera**, they all have a venom gland but it has become vestigial in non venomous species depending on evolutionary pressure (all the mostly venomous squamates also share a common ancestor). Also the Komodo dragon and some other monitors have venoms.)



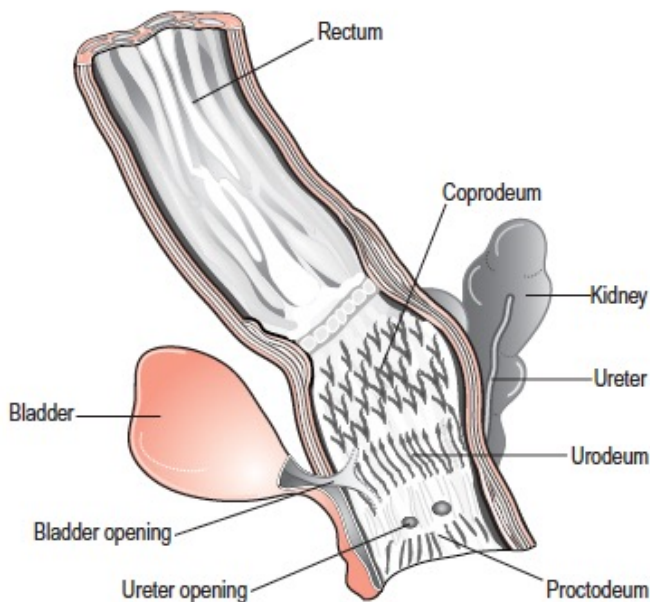
The tongue is forked in snakes and some lizards such

as monitor lizards. The forked tongue allows to gather more scent molecules during flickering, but also to know where scents come from (directional smelling in a way). At rest, the tip goes in the **Jacobson' organ (also called the vomeronasal organ)** on the roof of the mouth. In chameleons, the tongue is projective, sticky, and prehensile in most species.



While reptile metabolism is slow, a digesting snake can increase its metabolism by 7 to 10-fold (their heart more than doubles in size too during the process).

The cloacal organization is very similar to birds except that many reptiles have a bladder (see below). The phallus and opening of the scent glands are located in the proctodeum. Also the ureters do not open in the bladder, but in the urodeum and the urine then flows back into the bladder.



The liver is made of two large lobes, with the right typically bigger than the left. The liver is particularly prominent in chelonians and occupy most of the cranial coelomic cavity. Black areas in the liver are typically dense in melanomacrophages. The gallbladder is located caudal to the liver in snakes. Most reptiles also have a **hepatic portal system** that drains part of the caudal body. This has great clinical significance as drugs injected in the caudal half of reptiles will undergo first-

pass hepatic metabolism (so drugs primarily metabolized by the liver won't be as effective, exception is the ventral tail vein, which is fine to inject in).

The spleen is a small spherical organ present between the gallbladder and pancreas. In some species, it is tightly adherent to the pancreas and can even somewhat fused with it to form the splenopancreas (mainly in snakes).

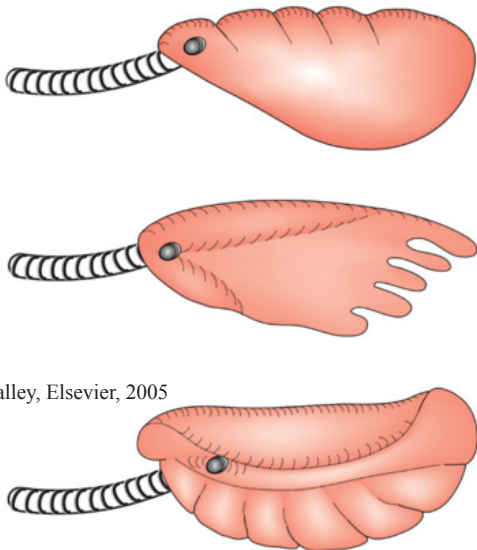
RESPIRATORY SYSTEM

The anatomy of the respiratory system is highly variable and of great clinical significance. The trachea has **complete tracheal rings in chelonians and crocodilians (like birds, remember they are closer to birds than other reptiles) and incomplete in squamates (lizards and snakes)**. The tracheal bifurcation is very rostral in chelonians so they should not be intubated too far. The tracheal diameter is much smaller than mammals and birds because of their lower metabolism and thus need for oxygen. Therefore, reptiles will need very tiny endotracheal tubes compared to other animals. The larynx is roughly similar to birds and is particularly mobile in snakes so they can breathe while swallowing a prey (which can take a while). As most reptiles lack a vocal structure (such as the syrinx in birds or the vocal cords in mammals), they are **mostly mute** (with a few exceptions, notably in crocodilians) or vocalize by hissing and expelling air quickly.



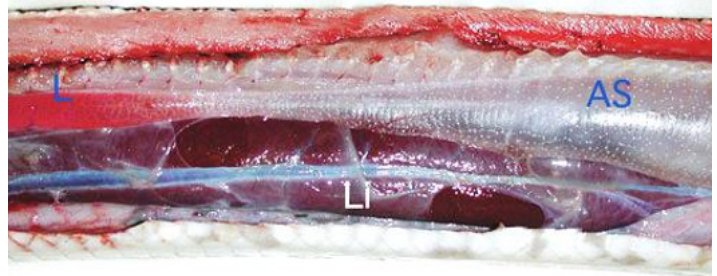
The variability in lung structure can be summarized by 3 big plans depending on the level of compartmentalization. **Unicameral lungs** only have one chamber and are found mostly in snakes (which makes it great for pulmonoscopy) and primitive lizards such as geckos

and skinks. **Paucicameral lungs** have a more complex organization but compartmentalization is still minimal; these are characteristic of most squamates such as iguanids, agamids, and chameleons. Finally, **multicameral lungs** with multiple chambers are found in advanced lizards such as monitor lizards (the most evolved squamates with the snakes), chelonians, and crocodilians. There are many exceptions to these rules as well. For instance some snakes have multicameral lungs. The lungs are mostly free floating in the pleuroperitoneal cavity except in monitor lizards, chelonians, and crocodilians (see Coelomic Cavities).

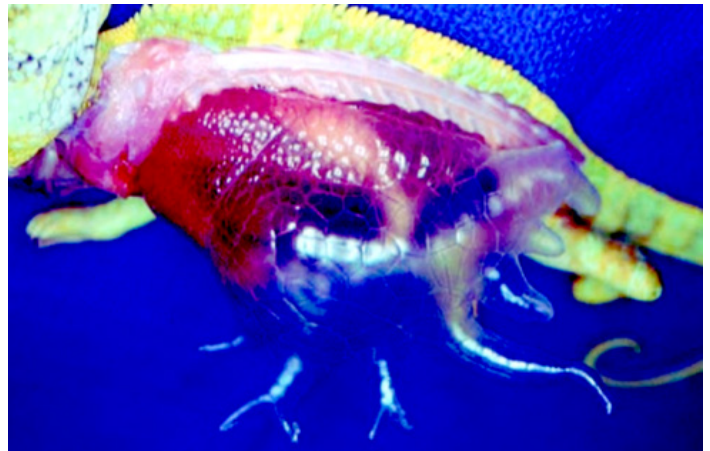


O'Malley, Elsevier, 2005

The gas exchange surface of reptiles is small (which fits with their metabolism) and is made of **ediculi** (wider than deep) or **faveoli** (deeper than wide), which are about 100x larger than mammalian alveoli. Since the exchange surface is so poorly developed, reptiles have to make up for it by having a higher volume, so reptiles have large lungs. **In chelonians, the ribcage is rigid and they lack intercostal muscles, so they have evolved a unique ventilatory mechanism using abdominal muscles** (mainly transverse abdominal which wraps around the lung for exhalation and oblique abdominal for inhalation, these muscles act by moving the post-pulmonary septum). Contractions of these muscles can move the legs and chelonians can also assist ventilation by retracting their head and legs within their shell. This is relevant because we incise this muscle during prefemoral celiotomy. **In squamates, the lung is divided into 2 parts: the cranial respiratory part made of pulmonary parenchyma and the caudal avascular part, which resembles avian air sacs.**



The air sac part is particularly extensive in snakes, skinks, and chameleons. In the latter, many tentacular diverticuli are present throughout the caudo-ventral part of their coelom.



Most snakes only have the **right lung** with the exceptions of primitive snakes such as boidae and pythonidae (the most common pet snakes basically), which have 2 lungs. Some species also have accessory lungs such as the tracheal and cardiac lung of some snakes and chameleons. The tracheal lung of snake is basically just an expansion of the tracheal membrane to offer extra respiratory surface and it may appear undistinguishable from the right lung. **The snake respiratory clearance mechanism (poorly ciliated respiratory epithelium) is rudimentary** and makes the elimination of foreign particles or inflammatory exudates from the trachea or lungs difficult.

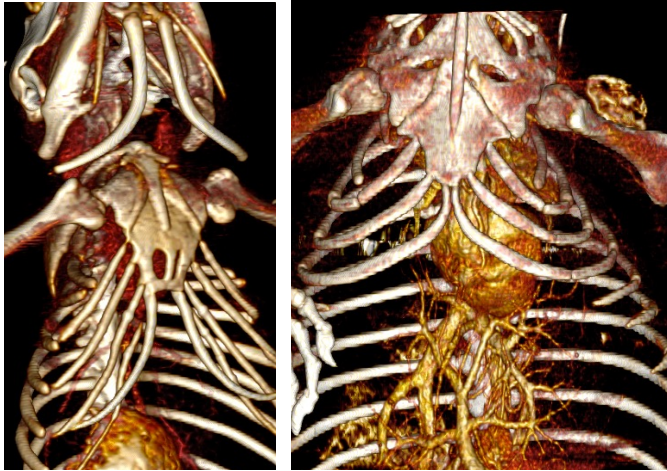
Some reptiles also use non-pulmonary respiratory surfaces such as the skin in soft-shell turtles, the pharyngeal mucosa in many squamates and chelonians, and the cloaca in freshwater turtles (especially during overwintering).

CARDIOVASCULAR SYSTEM

We will only discuss the heart of non-crocodilian species (crocodilians have a four-chamber heart with some communications remaining between the systemic and

pulmonary circulations).

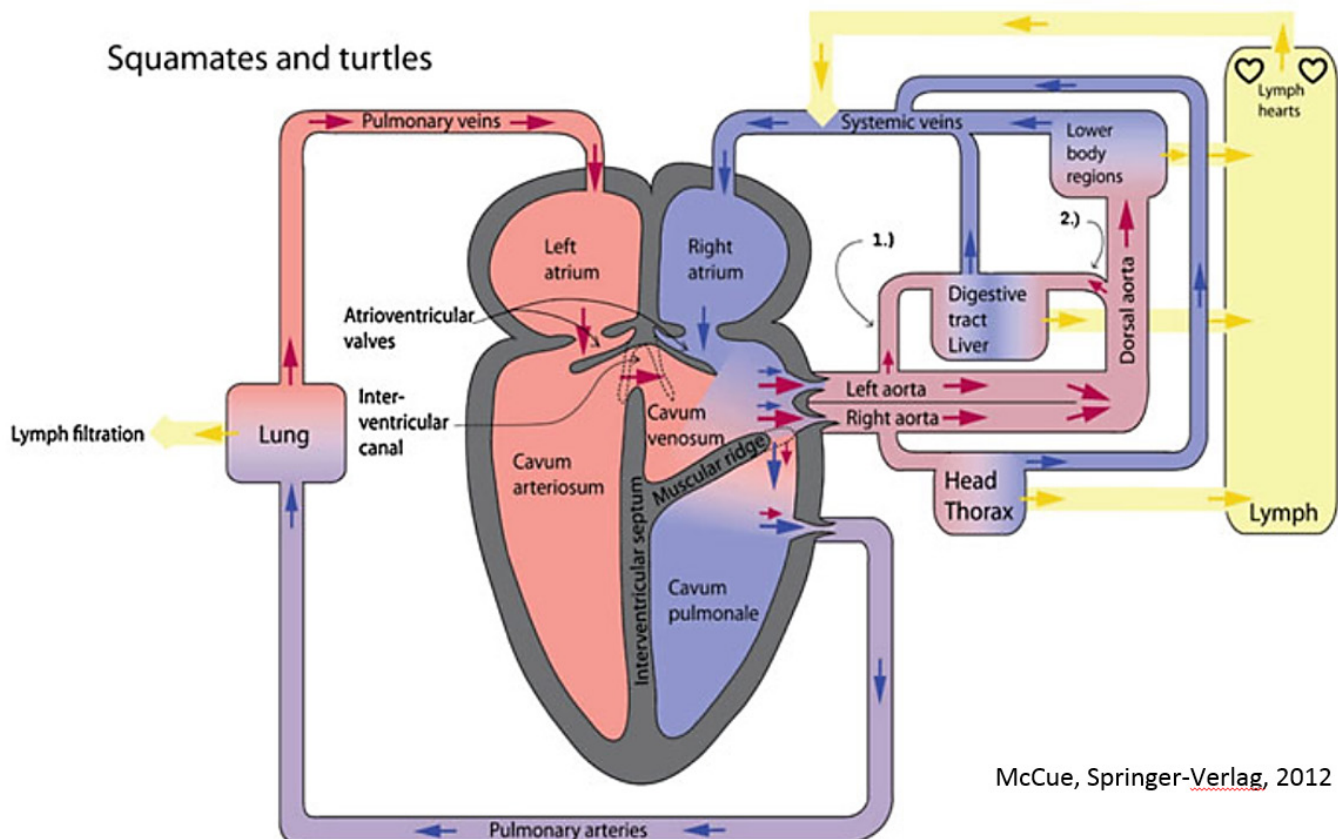
The heart is located fairly cranially in most reptiles. It is **at the level of thoracic girdle** in most lizards, except in monitor lizards in which it is slightly more caudal. In snakes, it can be found at the end of the caudal 1/4 - 1/3 of the body. Look at these 3D CT below (left is a bearded dragon, right is a monitor lizard, contrast was injected into the vascular system).



The heart is made of 2 atria and a single ventricle. However, the ventricle effectively functions as a 2 ventricular structure with still a good separation between the systemic and pulmonary circulation with limited blood mixing. **The single ventricle is three-dimensionally complex and made of 3 sub-cavities called**

the cavum arteriosum (homologous to the left ventricle in mammals and birds), the cavum venosum, and the cavum pulmonale (homologous to the right ventricle). There are 2 aortae both leaving the cavum venosum. Contractions are complex and basically work as follow (sit down for that): during early ventricular diastole, the cavum venosum and pulmonale receive deoxygenated blood from the right atrium and the cavum arteriosum receives blood from the left atrium. During late diastole, blood flow from the atria has ceased and deoxygenated blood from the cavum venosum flows into the cavum pulmonale, the atrioventricular valves close, which opens the interventricular canal and allows blood from the cavum arteriosum into the cavum venosum. During late ventricular systole, the muscular ridge separates the cavum venosum from the cavum pulmonale. Blood from the cavum pulmonale is ejected into the pulmonary artery and blood from the cavum venosum is ejected into the aortae. You got it? no! Well look at that diagram below then and see if it makes sense.

The important thing about that whole system is that reptiles can have a **right to left shunt** that is physiological and regulated as needed. The shunt is regulated by the differential of pressure between the aortae and the pulmonary arteries. If a reptile does not need to breathe (such as when diving, or breath holding), it will shunt blood away from the lungs (right to left shunt), which

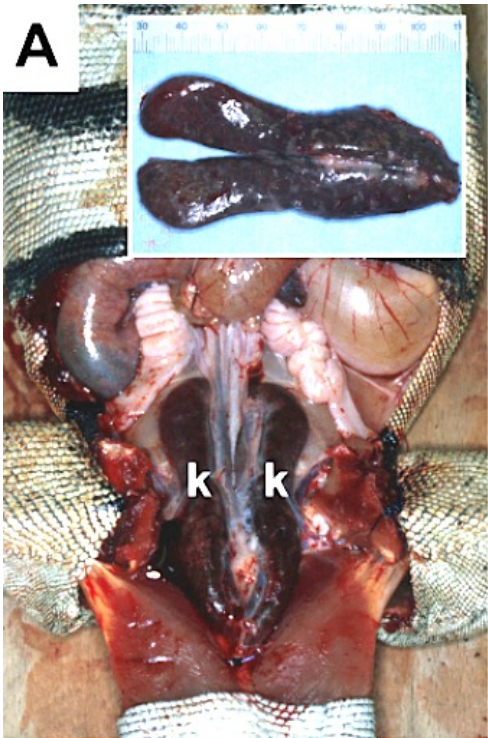


may be annoying when trying to mask-induced them or give them oxygen (also if they shunt because of a high oxygen flow for instance, they may not get rid of their isoflurane and can take longer to recover). *Interesting note: in clinics you may try to avoid shunting when recovering reptiles with room air and also by giving them a shot of atropine, also dexmedetomidine may increase shunting (I let you ponder that one).*

Cardiac shunting is favored by evolution in many species so it is not necessarily a primitive feature as it allows reptiles to adapt to specific niches. For instance, crocodilians have evolved a 4 chamber heart, but have also developed a new way to perform cardiac shunting because it is so useful for their digestion and their aquatic lifestyle.

RENAL SYSTEM

The reptilian kidney is relatively simple and mostly made of **loopless nephrons**. Therefore, their urine is not concentrated, but can get modified by the **post-renal handling of urine** (much like in birds, or should I say birds are like reptiles). The kidneys are located dorsal and caudal. In some species, the kidneys are very caudal and may extend within the tail base in lizards (so you can take a renal biopsy by a lateral incision in the tail).



Male snakes have a caudal portion of the kidneys that is enlarged and called the **sexual segment** (produces seminal fluid, does not misdiagnose for a tumour).

A urinary bladder is present in chelonians and most lizards (but not all, it would be too simple!). In terrestrial tortoises, they use their bladder to store water and they can reabsorb water and electrolytes through the bladder wall. **Snakes and crocodilians do not have a bladder.** The ureters open into the cloacal urodeum and not the bladder (which means that you can remove the bladder surgically if needed in reptiles).

Most reptiles are **uricotelic** like birds. However, aquatic reptiles, which have unlimited access to water may have a simpler strategy (uric acid is costly to produce) and may produce significant ammonia and urea. It is mainly true in chelonians (freshwater and sea turtles). Terrestrial chelonians also excrete significant urea as well as the ancient tuatara.

Species	% of total urinary nitrogen		
	Ammonia	Urea	Uric acid
Testudines			
Wholly aquatic	20-25	20-25	5
Semi-aquatic	6-15	40-60	5
Wholly terrestrial			
Moist environment	6	30	7
Dry environment	5	10-20	50-60
Crocodilians	25	0-5	70
Squamates			
Lizards	little	0-8	90
Snakes	little	0-2	98
Rynchocephala	3-4	10-28	65-80

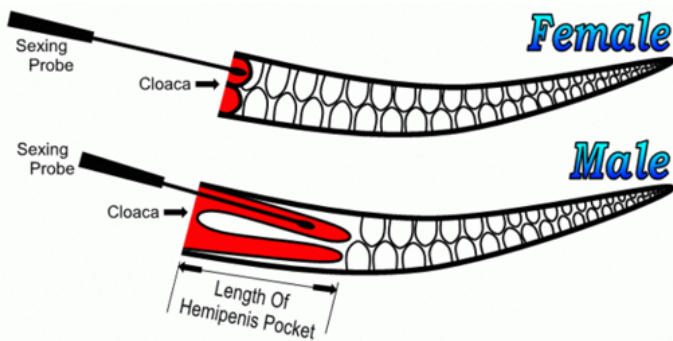
Consequently, it is important to have the urea on the biochemistry panel for these species. On biochemistry, the **calcium-to-phosphorus ratio** is actually the most useful analyte compared to urea and uric acid for evaluating reptilian renal function.

Reptiles (and birds) also have a renal portal system, which drains part of the caudal body, and perfuses the renal tubules.

REPRODUCTIVE SYSTEM

The gonads, testis or ovaries, are intracoelomic in reptiles, much like in birds. When doing surgery, it is important to remember that **the right gonad is close to the vena cava and the left close to the adrenal gland**. Chelonians and crocodilians have a **phallus**. Squamates have **hemipenes** that are inverted into a socket caudal to the cloaca. **In snakes, the hemipenial sockets can be probed for sexing (deeper in males, you rarely go deeper than 2-4 scales in females).** In lizards, you

may infer the presence of the hemipenes by observing **hemipenal bulges** caudal to the cloaca. Hemipenes are calcified in monitors (so you can sex them by just taking an xray).



Most reptiles lay eggs like birds (and monotremes, but this is another story). Some reptiles are **viviparous** and have some types of placental exchange. Viviparous reptiles include some chameleons (Jackson's chameleon), boids, and viperids. A few reptiles are **ovoviviparous** (eggs hatch before being laid).

Some interesting facts that will keep you up at night include:

- Parthenogenesis** in some squamates (asexual reproduction in females, particular form of meiosis).

- Sex ratios in some species that have not evolved sex chromosomes (heteromorphic chromosomes) can be **regulated by temperature** (which is a bummer with climate change). Those include many species of turtles, all crocodilians, and a few squamates. The bearded dragon has heteromorphic chromosomes, but very high temperature can provide a trigger that override this to produce more females.

- Most reptiles **do not incubate their eggs** (except crocodilians and some snakes) and **cannot put pigments on their egg shells** (so they are all white whereas avian eggs have species-specific patterns and can be very colourful (adaptation to avoid brood parasitism))

Most reptiles can be sexed morphologically when adults, it does not tend to work as well in juveniles. In chelonians, males have a **concave plastron** (so they can mount the females), **longer toe nails** (for grasping the female), and **caudally placed vent** on the tail (so it can go under the female plastron for copulation). Some tortoises may also have other characteristics such as the chin glands and gular horns in California desert tortoise or wider anal scutes in Sulcata tortoise. In lizards, most of the males have prominent **femoral pores** (see in integument section) and you may see the **hemipenal bulges** caudal to the cloaca. Some species are also sexually dimorphic such as the green iguana where the

male is much bigger, kind of yellowish, and big headed (typical guys!!).

Geckos tend to have more precloacal pores in males.

NEUROLOGICAL SYSTEM

The nervous system is similar in organization than in birds with a **nuclear brain organization** with a pallium and spinal cord that follows vertebral segments (and not more cranial like in mammals and with no cauda equina). There are **no intervertebral disks** in reptiles. As reflexes are temperature dependent, you need to make sure reptiles are warmed up before conducting a neurological examination. Interestingly, the sidewalls of the braincase are cartilaginous in lizards and completely absent in snakes.

Reptiles have many "weird" behaviours (weird is the new normal!) that can be misinterpreted as neurological abnormalities. For instance, hognose snakes will feign death, be on their back with the tongue out and excrete a foul smelling substance (the neurological examination will obviously be a little bit tricky, sneaky snake!).

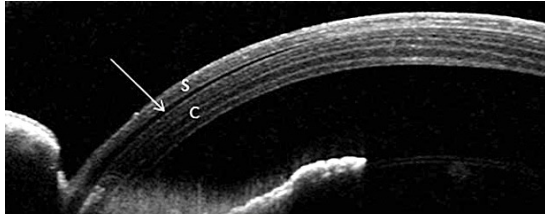


Bearded dragons do hand circumduction (like they are saying goodbye) to communicate with conspecifics. The list goes on and on (and on).



SPECIAL SENSES

The eyes are similar to birds, but less advanced. Snakes and most geckos (except leopard geckos) do not have open eyelids, they are fused to form a transparent structure called the **spectacle** that is shed during ecdysis (a spectacular process in them!!).



Most lizards have a third eye on top of their head that is called the **parietal** eye (has vestigial cornea and retina) that is connected to the pineal gland and probably helps them sense photoperiod and amount of sun/UV exposure.

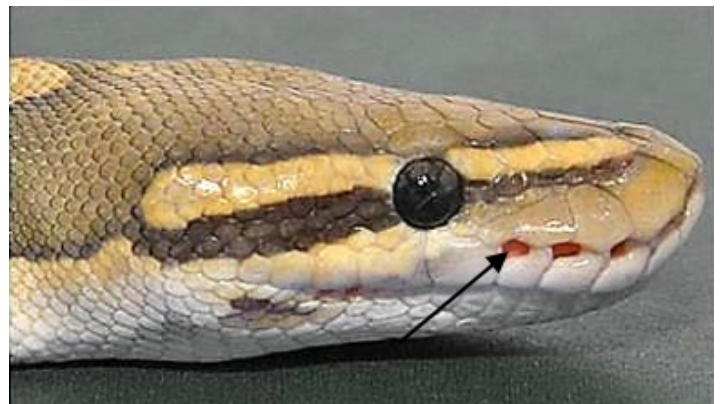


Most reptiles have a **Jacobson organ** (=vomeronasal organ) located on the roof of the mouth rostrally. The tip of the tongue goes in there and it is part of the sense of smell, which is hundreds of times better in snakes than in people.



Snakes, chameleons, and tuataras do not have external and middle ears and may sense sounds through vibration of the ground or branches.

Many snakes have **infrared receptors** around their lips (lower or upper, a single hole in pit vipers, several in other snakes, often called pit organs). As such, snakes can see their homeothermic unsuspected meek preys in the dark.



The Ten Most Important Facts

1. There are 4 orders of reptiles, 2 that are commonly seen in practice (Testudines and Squamates, most reptiles being squamates). Reptiles are very diverse, which manifests by great variability in anatomy and physiology.
2. Reptiles have a slow metabolism, they are poikilothermic and ectothermic.
3. Reptiles integument presents major differences from mammals in that reptiles shed their skin periodically in a process called ecdysis and numerous species have osteoderms. Chelonians have a shell divided into the carapace, the plastron, and the bridges.
4. Most reptiles have a single pleuroperitoneal cavity where all the organs are free floating. A few exceptions exist such as the crocodilians, chelonians, monitor lizards, and tegus.
5. Reptiles can have acrodont, pleurodont, or thecodont dentition. These differences have clinical implications, especially on the rate of periodontal disease.
6. The respiratory system is simple with unicameral, paucicameral, or multicameral lungs, depending on the species. Chelonians and crocodilians have complete tracheal rings whereas squamates have incomplete tracheal rings.
7. While the heart has a single ventricle, it functions as 2 (with minimum blood mixing) because of its complex three-dimensional anatomy with 3 sub-chambers and a complex contraction cycle. Cardiac shunting occurs in reptiles depending on physiological and ecological conditions.
8. Reptiles have simple kidneys that are located dorsally and caudally within the pelvis, that are composed of loopless nephrons. Most reptiles are uricotelic, but aquatic reptiles and chelonians have some degree of ammonotelism and ureotelism.
9. Chelonians and crocodilians have an intracloacal phallus and squamates have hemipenes. Lizards can be sexed by looking at hemipenal bulges in males and snakes can be sexed by probing of the hemipenal sockets.
10. The external ear is mainly composed of a visible tympanic membrane. Snakes, chameleons and tuataras do not have external or middle ears.